

NAKUUL AGARWAAL

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Full Working Rights (Student Visa Subclass 500)

WORK EXPERIENCE

University Of New South Wales (UNSW), Sydney, Australia

Dec 2025 – Present

IT Infrastructure Assistant, Technology Delivery

- Deliver first-level IT infrastructure support for university network services, to maintain operational continuity across campus.
- Assist in service activation, testing, maintenance and incident response.
- Perform initial fault diagnosis, escalating issues as required to meet service priorities and response timeframes.
- Support infrastructure upkeep through site audits, equipment checks, upgrades and infrastructure maintenance.

IAMOPS, Pune, India

Jan 2025 – Jul 2025

DevOps Engineer Intern

- Gained hands-on experience with AWS and GCP services, understanding cloud infrastructure and deployment.
- Worked on S3 bucket management, EC2 instance setup, and Lambda functions for server less computing.
- Collaborated with the DevOps and QA teams, working closely with clients to meet project requirements.
- Managed and guided interns, overseeing tasks and generating reports to track progress.

Ajeenkya DY Patil University, Pune, India

Apr 2024 – May 2025

Lead Placement Coordinator

- Managed the placement process, ensuring smooth recruitment for students and companies.
- Handled POD.AI application to post campus recruitment drives.
- Shared student data with recruiters and coordinated interviews.
- Organised on-campus drives and streamlined the hiring process.

Google DSC, Pune, India

Aug 2022 – Oct 2023

Management Lead

- Conducting various sessions and workshops on various technologies including Kotlin, AI, ML in our university impacting 100+ students and helping them learn newer modern technologies.
- Headed the first and only technology-focused club in my university with the aim to create a complete culture shift in the student community towards peer learning and embracing the community culture.
- Led a team of 10 Leads that architected and accomplished goals.

Startup Pedia, India

Jan 2022 – Jun 2022

Graphic Designer

- Designed branding materials and social media graphics.
- Created and managed Instagram posts to boost engagement.
- Collaborated with the marketing team to enhance digital presence.

The Sparks Foundation, Pune, India

Oct 2021 – Mar 2022

Intern

- Worked on projects related to web development, data science, and community engagement.
- Assisted in building applications and features for social impact.
- Gained hands-on experience in working with real-world data and problem-solving.

Vodafone Strategic Business Case Study (*Telecom Strategy, Revenue Model, Customer Experience*) ([Detail Website](#))

- Conducted a case study on Vodafone's telecom business structure, with focus on customer offerings, service-based revenue streams and digital engagement strategy.
- Evaluated the financial structure through customer plans, service monetisation, retention potential and operating efficiency as key drivers of business performance.
- Highlighted strategic upside through improved customer adoption, service monetisation, retention strength and scalable growth across core telecom operations.

HSBC Strategic Business Case Study (*Banking Strategy, Financial Structure, Digital Services*) ([Detail Website](#))

- Developed a case study on HSBC's banking and financial services structure, with emphasis on income streams, customer banking solutions and digital service expansion.
- Analysed the role of technology investment, cloud-based risk infrastructure and digital transformation in improving service delivery and operational effectiveness.
- Identified potential gains in terms of customer value, process efficiency, risk responsiveness and sustained competitive advantage in a digital-first banking environment.

AI-ML based intelligent de-smoking/de-hazing algorithm (*Deep Learning Arch., GANs, Segmentation, Dataset and Training*)

- Designed an AI model to **remove smoke and haze** from images and videos, enhancing visibility for rescue operations.
- Utilised **GANs, CNNs, and real-time processing algorithms** to reconstruct clear images.
- Applied deep learning models such as **CycleGAN, Pix2Pix, and DehazeNet** for improved accuracy.

Sign Language Translation Using Hand Detection (*Open CV, Image Processing*)

- Developed a **hand gesture recognition system** using **OpenCV and Mediapipe** to assist the deaf and mute in communication.
- Implemented **hand skeleton tracking** and a **rule-based classifier** to detect and interpret finger movements.
- Improved key point accuracy for better gesture recognition.

ML-Based Data Download Duplication Alert System (*Machine Learning, Anomaly Detection, Data Security*)

- Built an **anomaly detection system** to prevent redundant data downloads in cloud platforms.
- Used **Random Forest, Isolation Forest, DBSCAN, and LSTM-based models** to flag suspicious activities in real-time.
- Helped optimise **storage costs, data security, and download efficiency**.

Predictive Analytics for Stock Market Trends (*AI, Machine Learning, Financial Forecasting*)

- Developed an **AI-powered stock market prediction system** using historical data and sentiment analysis.
- Implemented **LSTM, ARIMA, Random Forest, and XGBoost** for accurate trend forecasting.
- Assisted investors in making **data-driven decisions and optimising investment strategies**.

Fire Detector (*IR diode*)

- Developed a **fire detection system** using **infrared (IR) diodes** to quickly identify fire incidents.
- Designed to **detect heat signatures and flame intensity** in real time, triggering alerts to prevent potential hazards.
- Utilised **sensor-based automation** to enhance **fire safety in residential and industrial spaces**.
- Helps in **early fire detection**, reducing the risk of **life loss and property damage**.
- Can be integrated with **IoT and alarm systems** for **automated emergency response**.

SKILLS

Technical Skills: C++, Python, AWS, GCP, Deep Learning, NLP, ML, Data Analytics, Git, Power BI

Soft Skills: Leadership, Team Management, Public Speaking, Problem-Solving, Time Management, Adaptability, Collaboration

CERTIFICATE

- **Oracle Cloud Infrastructure Foundations Associate – Oracle** *Issued Nov 2024*
Gained expertise in OCI services, networking, storage, and security principles.
- **AWS Cloud Practitioner – Amazon Web Services (AWS)** *Issued Sep 2023*
Developed a foundation knowledge in AWS cloud concepts, global infrastructure, and core services.
- **Natural Language Processing – DeepLearning.AI** *Issued Apr 2023*
Learned text processing, sentiment analysis, and language modelling using deep learning techniques.
- **Introduction to Artificial Intelligence (AI) – IBM** *Issued Dec 2022*
Understood AI fundamentals, machine learning algorithms, and real-world AI applications.
- **Google Cloud Platform – Google Developers Group** *Issued Oct 2022*
Acquired knowledge of Google Cloud services, cloud computing models, and deployment strategies.
- **Python for Everybody – University of Michigan** *Issued Sep 2022*
Strengthened Python programming skills, covering data structures, automation, and APIs.
- **Virtual Reality – Ajeenkya DY Patil University** *Issued Apr 2022*
Explored VR technology, immersive experiences, and human interaction in virtual environments.
- **Basics of Financial Markets – Elearnmarkets** *Issued Feb 2022*
Gained insights into financial instruments, stock markets, and investment principles.
- **Programming in C++ – Aptech** *Issued Jan 2019*
Developed proficiency in C++ programming, object-oriented concepts, and data structures.

VOLUNTEERING

- **National Cadet Corps (NCC) – Corporal**
Served as a Corporal in NCC, focusing on discipline, leadership, and national service. Engaged in various civil rights and social action programs, fostering teamwork and responsibility.
- **International Society for Krishna Consciousness (ISKCON) – Coordinator of Volunteers**
Managed and coordinated volunteer activities for social service initiatives. Led teams to organise community programs, events, and outreach activities.
- **Art of Living – Volunteer**
Actively participated in social service programs, promoting mental well-being, stress management, and community development.
- **National Service Scheme (NSS) – Student Volunteer**
Contributed to community service projects, participating in awareness campaigns, social welfare programs, and rural development activities.
- **Reap Benefit – Environmental Volunteer**
Engaged in environmental initiatives, working on sustainability projects to promote eco-friendly practices and climate awareness.

- **Smile Foundation – Fundraiser**

Worked towards child welfare and education by raising funds to support underprivileged children and their access to education and healthcare.

- **Ajeenkya DY Patil University – Event Organiser**

Organised cultural and technical events, managing logistics, planning, and execution of various university programs to enhance student engagement.

PUBLICATIONS

AI & ML-Based Data Duplication Alert System

Dec 18, 2024

This research paper introduces an intelligent system that helps detect and prevent unnecessary file downloads using machine learning. By keeping an eye on download requests, user actions, and related metadata, the system can spot patterns that suggest files are being downloaded more than once by mistake. It uses advanced techniques like anomaly detection, behaviour-based clustering, and predictive analysis, with models such as Isolation Forest, DBSCAN, LSTM, and decision tree ensembles, to tell the difference between intentional downloads and accidental repeats. This solution is especially useful for businesses, cloud storage providers, and data platforms looking to save bandwidth and maintain clean, efficient data systems.

EDUCATION

University Of New South Wales - Sydney, Australia

2028

Degree: Master of Information Technology And Commerce

Specialisation: Data Science & Engineering And Strategy & Innovation

Ajeenkya DY Patil University - Pune, India

2025

Degree: Bachelor of Technology in Computer Engineering

Specialisation: Artificial Intelligence And Machine Learning

CAREER OBJECTIVES

To be involved in an environment where I can utilise and apply my knowledge and skill and practical application with an opportunity to enhance my capabilities which effectively contribute to the growth of an organisation.